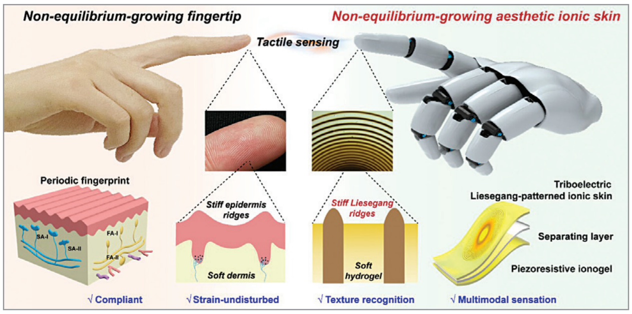
Ionic skin for robots with tactile abilities mimics human fingertips

Donghua University researchers in China have developed an ionic skin for robots to mimic human fingertips. By forming and characterising fingerprint patterns on soft material, researchers created a soft ionic skin with human-like tactile abilities. The skin mimics human fingertips with periodic ridges and valleys formed by a non-equilibrium reaction-diffusion process from the biochemical Turing effect. Fingertip ridges act as microlevers and maintain strain-free pressure sensitivity. The team created this ionic skin inspired by fingertip features using non-equilibrium Liesegang patterning in an elastic hydrogel. Liesegang patterning creates periodic precipitates, and a fluoroelastomer layer on the patterned skin creates a triboelectric tactile sensor that can detect pressure and recognise textures. The team combined the ionic skin with a piezoresistive ionogel to mimic a biological sensory system and conducted preliminary tests on the resulting sensor.



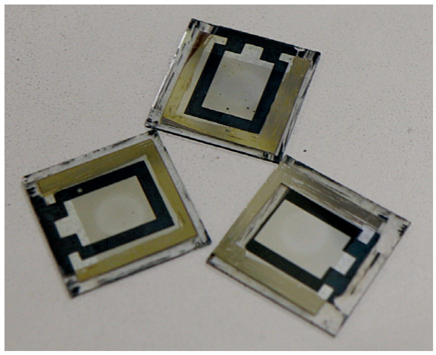
Non-equilibrium-growing aesthetic ionic skin for fingertip-like strain-undisturbed tactile sensation and texture recognition (Credit: Haiyan Qiao, Donghua University, China)

Atomically thin artificial neurons process light and electric signals for computing

Researchers at the University of Oxford, International Business Machines (IBM) Research Europe, and the University of Texas have created atomically thin artificial neurons that can process light and electric signals for computing. 2D materials have unique properties that can be adjusted by layering atoms. A device was made from graphene, molybdenum disulfide, and tungsten disulfide that changes conductance using light/electricity duration and power, unlike digital storage. It works like biological synapses, gradually changing electronic charge through electrical/optical signals to mimic neurons. This concept advances current artificial neural networks, with applications in AI hardware and neuromorphic engineering. It also advances our understanding and emulation of the brain. The researchers used a heterostructure stack to create semiconductor junctions and graphene electrodes that mimic actual neurons. This toolkit explores the potential of 2D materials in novel computing paradigms.

New additive can produce scalable, efficient perovskite solar cells

City University of Hong Kong researchers have developed a multifunctional and non-volatile additive to modulate perovskite film growth, improving the efficiency and stability of perovskite solar cells. Researchers developed a simple method to enhance perovskite film quality by adding a multifunctional molecule to the precursor, forming an intermediate phase. This regulated the crystallisation process, creating films with larger and more uniform crystal grains. The molecule also reduced non-radiative recombination losses, improving the overall film quality. 4-guanidinobenzoic acid hydrochloride led to lower defect density and increased power conversion efficiency in perovskite films, reaching 24.8% with reduced energy loss. The modified devices also showed improved thermal stability, with 98% of initial efficiency. The findings demonstrate potential for scalable, highly efficient perovskite solar cells.



The 1cm^2 perovskite solar cells with additive (Credit: City University of Hong Kong)

New technique can help create precise devices and extend their lifetimes

University of Melbourne researchers have simplified piezoelectric material benchmarking by incorporating electrostatic signals into signal processing techniques. Piezoelectric polymers convert motion into electricity for wearables but may build up electrostatic charges from friction. Conductive adhesive wrapping was used to differentiate intrinsic signals from hindered signals. Shielded harvesters showed a distinct frequency response, distinguishing them from unshielded ones. Researchers used fast Fourier transform to detect phantom energy in energy harvester electrical output. The distorted frequency spectrum of the signal, caused